

NO:

DATE:

Probability & Statistical Data Analysis

Assignment 1

Question 1

- a) Population : Students of Faculty of Computing
 Sample : A selected group of students from Faculty of Computing
- b) i. Discrete : number of respondent, number of courses failed
 Continuous : time spent on studying per day, household income, GPA result, age of respondent
- ii. Categorical : gender of respondent, ethnicity of respondent
 Numerical : GPA result, age of respondent, household income, time spent on studying per day, number of respondent, number of courses failed
- iii. Nominal : - Gender of respondent (Male / Female)
 - Ethnicity of respondent (Malay / Chinese / India...)
- Ratio : - Time spent on studying per day (1h, 2h, 3h...)
 - Number of respondent (30)
 - Number of courses failed (0, 1, 2...)
 - GPA result (0.0, 4.0...)
 - Age of respondent (21, 22...)
- Interval : - Household Income (0, 3000, 4000...)
- c) Primary Data Source
 ↳ All data from the survey
- d) Secondary Data Source
 ↳ Data from books and educational websites
- e) Possible statistics :
 - Mean (Average of GPA)
 - Median (Household income)
 - Mode (Time spent on studying)
 - Range (Household income)

Question 2

No. of students	Frequency	Relative frequency
0	3	0.10
1	2	0.07
2	7	0.23
3	7	0.23
4	5	0.17
5	4	0.13
6	2	0.07
Σ	30	1.00

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Question 3

(i)	Weight	Tabulation	Frequency	Weight	Tabulation	Frequency
	450		10	466		4
	451		2	467		5
	452		2	468		0
	453		7	469		4
	454		3	470		4
	455		6	471		9
	456		4	472		9
	457		6	473		1
	458		5	474		
	459		2	475		
	460		3	476		
	461		2	477		
	462		2	478		
	463		3	479		
	464		4	480		
	465		3			

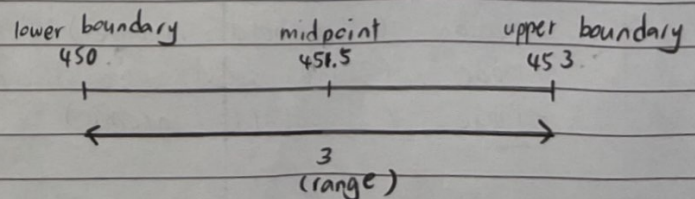
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(ii) Range = $473 - 450$
 $= 23$

(iv) $MP_1 = 450 + \frac{3}{2}$
 $= 451.5$

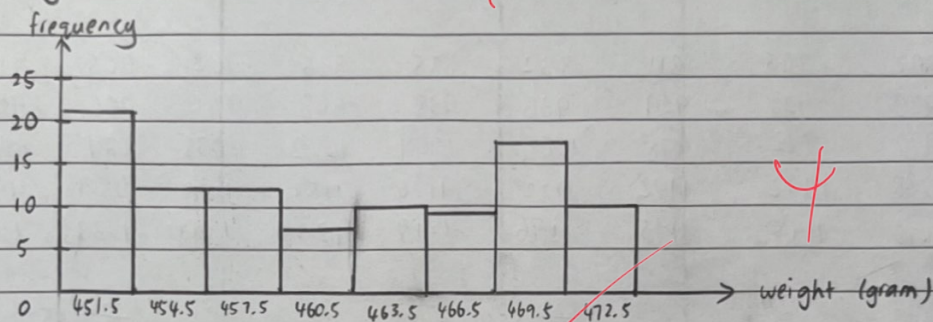
(iii) $i = \frac{23}{1 + 3.322 \log 100}$
 $= 3.00$

v Cell boundaries

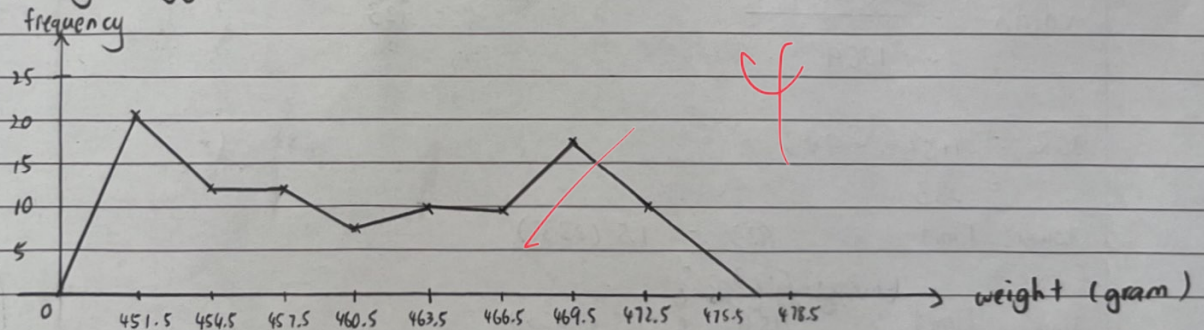


Cell boundaries	Cell midpoint	Frequency	Relative frequency (RF)	Cumulative (RF)
450 - 453	451.5	21	0.21	0.21
453 - 456	454.5	13	0.13	0.34
456 - 459	457.5	13	0.13	0.47
459 - 462	460.5	7	0.07	0.54
462 - 465	463.5	10	0.10	0.64
465 - 468	466.5	9	0.09	0.73
468 - 471	469.5	17	0.17	0.90
471 - 474	472.5	10	0.10	1.00

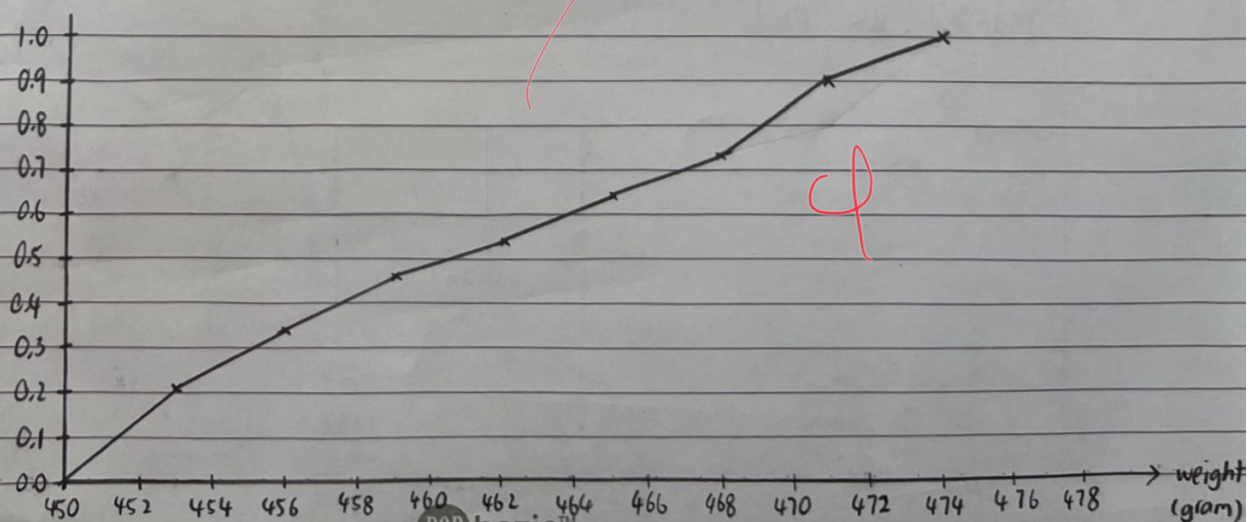
a) Histogram



b) Frequency Polygon



c) Ogive



Question 4

919	1196	78	1126	936	918
1156	920	948	1067	1092	1162
1170	929	950	905	972	1035
1045	855	1195	1195	1340	1122
938	970	1237	956	1102	1157
978	832	1009	1151	1151	1009
765	958	902	1022	1333	811
1211	1085	896	958	1311	1037
702	923				

78	702	765	811	832	855	896	902	905	918
919	920	923	929	936	938	948	950	956	958
958	970	972	978	1009	1009	1022	1035	1037	1045
1067	1085	1092	1102	1122	1126	1151	1156	1157	1157
1162	1170	1195	1195	1196	1217	1237	1311	1333	1340

$$\min = 78$$

$$\max = 1340$$

$$\frac{50}{2} = 25^{\text{th}}$$

$$\text{median} = \frac{25^{\text{th}} + 26^{\text{th}}}{2}$$

$$= 1009$$

$$\frac{25}{100} \times 50 = 12.5$$

$$\approx 13^{\text{th}}$$

$$Q1 = 13^{\text{th}}$$

$$= 923$$

$$\frac{75}{100} \times 50 = 37.5$$

$$\approx 38^{\text{th}}$$

$$Q3 = 38^{\text{th}}$$

$$= 1156$$

$$IQR = 1156 - 923$$

$$= 233$$

$$\text{Lower limit} = 923 - 1.5(233)$$

$$= 573.5$$

$$\text{Upper limit} = 1156 + 1.5(233)$$

$$= 1505.5$$

Modified Box Plot

outlier

